



LY CHINH TAN

Biomedical Engineering | Physiological Time Series | Deep Learning
Biosignal Analysis | Machine Learning

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Portfolio: lct-dotcom.github.io/LCT-Profile/ | GitHub: [LCT-dotcom](https://github.com/LCT-dotcom)

ResearchGate: researchgate.net/profile/Ly-Tan-2?ev=hdr_xprf

PROFILE

Fourth-year Biomedical Engineering student at HCMUT and first author of a published activity-recognition paper and a PPG manuscript under revision. Experience in wearable signal analysis, ECG domain adaptation and medical image processing, with an emphasis on reproducible experiments and subject-level evaluation. Currently developing embedded-systems knowledge through STM32 study.

EDUCATION

Ho Chi Minh City University of Technology, VNU-HCM

Engineering Physics, Biomedical Engineering major | August 2023–Present

English-taught program | Cumulative GPA: 3.1/4.0 | Semester 252 GPA: 3.7/4.0

RESEARCH

Human Behavior Recognition Using 2D Time-Series Data and Machine Learning Models

First author with Assoc. Prof. Quang Linh Huynh | Journal of Physics: Conference Series 3180, 012004 (2026)

DOI: [10.1088/1742-6596/3180/1/012004](https://doi.org/10.1088/1742-6596/3180/1/012004)

- Developed an eight-class human activity recognition pipeline from 17 COCO keypoints; normalized pose coordinates by torso length, handled missing values and segmented movement into 5-second windows (150 frames at 30 FPS).
- Engineered TSFEL statistical, temporal and spectral features alongside joint velocity, acceleration, limb angles and hand-to-face distances; removed redundant descriptors for compact tree-based classification.
- Trained and compared HistGradientBoosting, XGBoost and LightGBM under subject-level evaluation; examined per-class confusion and difficult participants to identify errors between similar seated and hand-to-mouth activities.
- Extended the research with training-only feature selection, nested subject-level tuning, Viterbi decoding and unseen-participant inference. The later five-subject LOSO evaluation achieved 73.31% accuracy and 76.39% macro F1 under a protocol distinct from the published paper.

A Lightweight Temporal Model for Heart Rate Estimation from Wrist PPG under Daily-Life Motion

First author with Tuan Kiet Pham, Duy Trung Do, Phi Phong Pham and Assoc. Prof. Quang Linh Huynh
BME11 manuscript submitted; under revision, decision pending | August 2025–Present

- Built a synchronized PPG–ACC processing pipeline at 64 Hz, using 8-second windows with a 2-second stride; incorporated accelerometer-derived motion context, spectral priors and signal-quality information.
- Designed and implemented a 1.014M-parameter PyTorch model combining depthwise-separable residual CNNs, spectral features, 12-window temporal attention and a three-expert mixture to estimate heart rate, uncertainty and artifact probability.
- Conducted strict 15-fold LOSO evaluation on 64,697 windows: raw macro MAE 4.307 ± 2.020 bpm and decoded MAE 4.278 ± 1.935 bpm. Evaluated Viterbi decoding, quality gating and rescue stages separately; the full pipeline obtained 4.491 bpm MAE.
- Retrained six ablation variants across all 15 folds and applied paired Wilcoxon tests with Holm correction. Removing the residual CNN increased raw MAE by 1.561 bpm ($p < 0.001$); removing temporal attention increased it by 1.133 bpm ($p = 0.002$).

WORK EXPERIENCE

Coordinator | NCKU Overseas Hub in Vietnam

May 2026–Present

Intern | PS Medical

June–August 2026 | Approximately two months

Company specializing in laser equipment for aesthetic applications.

SELECTED PROJECTS

AD8232 ECG Domain Adaptation | Internship project

- Evaluated a three-class MIT-BIH/SVDB-trained ECG classifier on a 30-minute single-lead AD8232 recording; audited ventricular ectopic beat failures, channel representation and temporal-context mismatch.
- Adapted the Stage-B bidirectional GRU with attention while freezing Stage A and calibration rules. Used raw temporal splitting, segment-local feature extraction and boundary-safe context windows.
- Improved macro F1 from 0.513 to 0.967 on 2,062 eligible beats under five-fold temporal out-of-fold evaluation. These are within-record results; some abnormal beats may be synthetic and independent-session validation is pending.

Medical Imaging Labs | DSA, CT and MRI

- Implemented DSA subtraction, registration and vessel enhancement; CT Radon projections, BP/FBP, SIRT/CGLS; and MRI DICOM/NIfTI loading, spatial metadata, slice visualization and image comparisons.
- Organized three reproducible notebooks with configurable data paths, synthetic demos and numerical regression checks; verified the revised workflows on supplied coursework data.

Jarvis Research Assistant | Personal project

- Built a local paper-discovery and knowledge-management tool with multi-source retrieval, DOI/title deduplication, ranking, SQLite persistence, report export and optional Vietnamese summaries.
- Implemented a browser/CLI interface and a linked knowledge library with similarity search and clustering. Verified 91 offline checks and 16 end-to-end fixture checks.

TECHNICAL SKILLS

- Programming & ML: Python, PyTorch, scikit-learn, TSFEL, NumPy, pandas, SciPy, Jupyter and Git.
- Biosignals & Medical Imaging: PPG/ECG and physiological time-series analysis; OpenCV, scikit-image, pydicom and NiBabel; DSA, CT and MRI workflows.
- Embedded systems: Foundational STM32 programming and peripheral knowledge: GPIO, EXTI/NVIC interrupts, timers/PWM, UART/USART, I²C, SPI, ADC and DMA; introductory concepts of DAC, CAN and USB. Able to interpret peripheral registers and device reference manuals.
- Engineering tools: MATLAB, Arduino, AutoCAD and Raspberry Pi; reading technical drawings, component datasheets and medical-equipment documentation in English.

CERTIFICATES AND RECOGNITION

- ISAS Challenge 2025: 6th place, competing individually.
- BME11 2026: Best Presenter
- OISP Scholarship: Semesters 231 and 251.
- First Prize, 2023 Engineering Festival, for a PPG heart-rate monitoring sensor project of Faculty of Applied Science.

Neural Networks and Deep Learning — DeepLearning.AI | September 11, 2026

Google AI Essentials — Google | September 3, 2026 | Five-course specialization

IELTS 6.5 | MOS Excel 2019